Randy J. Nelson



An Introduction to Behavioral Endocrinology

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The Study of Behavioral Endocrinology

Does anabolic steroid abuse cause violent behavior? Does the sex drive wane with aging? Is the sex drive of women higher or lower than the sex drive of men? Can a woman's mood be affected by her menstrual cycle? Do seasonal cycles of depression occur in people? Why are more men than women in prison for committing violent crimes? Does postpartum depression really exist? Is homosexuality caused by hormone levels that are too low or too high? Is the sexual behavior of women influenced by menopause? Many of us have pondered these questions during informal discussions with friends and family members. Researchers in the field of behavioral endocrinology, the study of the interaction between hormones and behavior, attempt to address these kinds of questions in a formal, scientific manner.

The study of the interaction between hormones and behavior has been truly interdisciplinary: methods and techniques from one scientific discipline have been borrowed and refined by researchers in other fields. Anatomists, psychologists, entomologists, zoologists, physiologists, endocrinologists, and psychiatrists have all contributed to the understanding of hormone–behavior interactions. This exciting commingling of scientific interests and approaches, with its ongoing synthesis of knowledge, has led to the emergence of behavioral endocrinology as a distinct field of study (Beach, 1975).

Historical Roots of Behavioral Endocrinology

Ebbinghaus (1908) stated that psychology has a short history but a long past, and the same can be said of behavioral endocrinology (Beach, 1974a). While

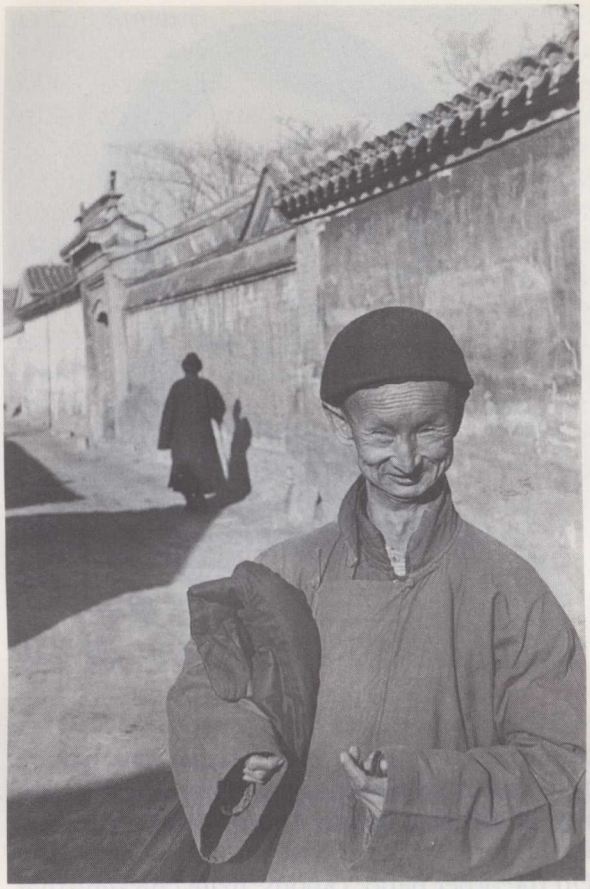
the modern era of the discipline began only about 50 years ago, some effects of hormones on behavior have been implicitly recognized for centuries.

The male sex organs, or testes, produce and secrete a hormone called testosterone, which affects sexual behavior, aggressiveness, territoriality, hibernation, and migration, as well as many other behaviors that differentiate males from females. The testes are usually located outside of the body cavity in mammals and can easily be damaged or removed. Thus, castration, the surgical removal of the testes, has historically been the most common manipulation of the endocrine system. For millenia many species of domestic animals have been castrated to make them better to eat or easier to control, and the behavioral and physical effects of castration have been known since antiquity. Indeed, these effects were known to Aristotle, who described the effects of castration in roosters with great detail and accuracy.

For centuries, royalty employed men castrated before puberty, or eunuchs, to guard women from other men. For example, the Old Testament reports that these emasculated males were used to guard the women's quarters of Hebrew kings and princes (Esther 1:10). Castration in humans often has little or no effect on physical appearance or future sexual behavior when performed after the unfortunate individual attains sexual maturation; however, if human males are castrated before puberty, they will develop a characteristic physical appearance, marked by short stature and long arms (Figure 1.1), and sexual behaviors are unlikely to develop. The typical secondary male sex characters are also affected by removal of the testes. For example, eunuchs never develop beards, and the pubertal change in voice does not occur. Normally during puberty, the vocal cords of males thicken in response to testosterone secreted by the testes. It is the thickened vocal cords that produce the deeper-pitched voice characteristic of males, just as the thick strings of a guitar produce deeper-pitched notes than the thin strings.

You may be surprised to learn that it was once a common practice in Europe and Asia to castrate young boys with exceptional singing voices to prevent the pubertal changes in pitch. These singers became known as *castrati*. Although castrati were prized by church choirs for centuries, their popularity reached a peak in Europe during the 17th and 18th centuries with the development of opera, which made castrati the first superstars of the entertainment world (Heriot, 1974). The first castrato opera star, Baldassare Ferri, died in 1680 at the age of 70 with a fortune worth the equivalent of $3,000,000 today. In hopes of attaining this level of wealth and fame, young boys with musical aptitude were identified early, and poor families offered their sons outright to church leaders, singing teachers, and music academies. Thousands of boys lost their testes but never gained the celebrity or riches of the star castrati.

What did a castrato sound like? Essentially, the castrati had the range of a soprano, but the greater development of the male lungs gave their singing remarkable power. An early critic remarked, "Their timbre is as clear and piercing as that of choirboys and much more powerful; they appear to sing an octave above the natural voice of women. Their voices . . .



1 ***EUNUCH OF THE LAST IMPERIAL COURT,*** photographed in China by Henri Cartier-Bresson in 1949. Note the lack of facial hair and unusually long arms. One eunuch of the Ching Dynasty, Sun Yaoting, still lives in China and is over 90 years old.

are brilliant, light, full of sparkle, very loud, and astound with a very wide range" (Heriot, 1974).

After 200 years, the tastes of the opera-loving public changed. The rise in popularity of the female soprano voice reduced the demand for the castrati, and they soon became an oddity. In 1849, the last great castrato, Giovanni Velluti, retired from opera to his villa in Venice.

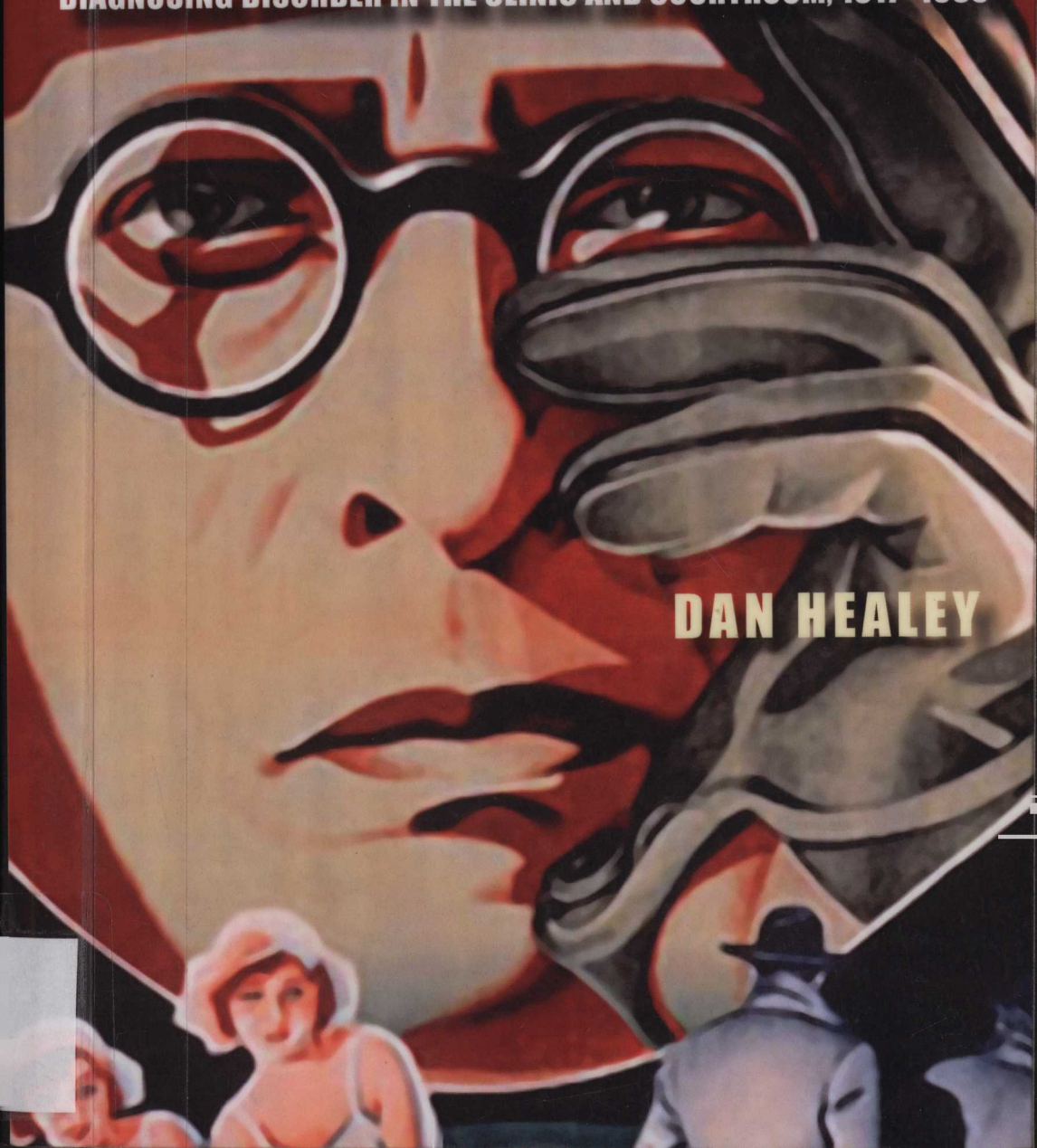
Berthold's Experiment

A useful starting point for understanding research in hormones and behavior is a classic 19th-century experiment that is now considered to be the first formal study of endocrinology. This remarkable experiment conclusively demonstrated that a substance produced by the testes could travel through the bloodstream and eventually affect behavior. Professor Arnold Adolph Berthold, a Swiss–German physician and professor of physiology at the University of Göttingen (Figure 1.2), used chickens in his experiment, and showed that a product of the testes was necessary for a cockerel (an immature male chicken) to develop into a normal adult rooster.

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BOLSHEVIK SEXUAL FORENSICS

DIAGNOSING DISORDER IN THE CLINIC AND COURTROOM, 1917–1939



DAN HEALEY

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INTRODUCTION

BOLSHEVIK MEDICINE AND RUSSIA'S "SEXUAL REVOLUTION"

That Russia experienced a "sexual revolution" in the wake of the Bolshevik political revolution of 1917 is considered axiomatic. Radical measures were adopted—some within weeks of the October seizure of power—to secularize marriage, to make divorce easy to obtain, to communalize childcare and housework, and later to legalize abortion on demand and to decriminalize male homosexuality. Vladimir Lenin and his radical socialists implemented a program that had roots not only in the European socialist tradition, but also in developments in Russia's liberal, professional, and intellectual circles, where impulses to increase the authority of law and medicine, at the expense of the tsarist autocracy, were strong.

"Modern" ideas about sexuality—the emancipation of women, the end or renovation of religious morality, the search for a new morality as part of a fairer social and political future, the application of apparently rational and scientific criteria to the sexual sphere—were hotly contested in tsarist debates about public values. In late Imperial Russia, doctors, lawyers, writers, social critics, and civic leaders who were engaged in these debates all agreed that a modern nation needed new forms of sexual regulation.[1] Whatever their differences, most commentators held science in particular esteem. The medical profession, including forensic experts, rapidly acquired cultural authority within and beyond the clinic, lecture hall, and courtroom, as purveyors of a fresh and convincing language of sexuality. Even elements in Russia's predominantly conservative Orthodox Church saw the value of "straightforward scientific talk rather than moral fables" in sex education.[2] The tsarist regime, marked by political paralysis, did little however to overhaul sexual regulation.

The short-lived liberal Provisional Government of February to October 1917 managed, in the realm of sexual regulations, to extend to women

the franchise before collapsing in the face of war and political pressure. Coming to power on the heels of this collapse, the Bolsheviks acted as condensers for the widespread impulse to sexual modernity, when in their earliest declarations and later lawmaking, they implemented radical policies. These early declarations remained largely aspirational until the crisis of the Civil War (1918–1921) receded. The era of mixed-market economics, the New Economic Policy (NEP, 1921–1928), saw Bolsheviks and fellow-traveling professionals construct new medical and legal institutions that would give expression to sexual modernity. Forensic medicine, located at the conjunction of law and medicine, was uniquely placed to interpret the sexual revolution that was about to be promoted by the new regime.

What did "revolution" mean for sexuality? The Bolsheviks arrived in power with received ideas rather than a worked-out strategy for "sexual revolution." Russia's radicals had absorbed the European heritage of socialist thinking on sexuality, and certain works translated into Russian, such as August Bebel's *Women and Socialism* (1879) and *The Origin of the Family: Private Property and the State* (1884) by Friedrich Engels, were regarded as "catechisms" by party thinkers and activists. Critiques of the patriarchal family and the enslavement of women inside church- and state-sanctioned matrimony, an institution supposedly designed merely to protect private property, circulated widely in late nineteenth-century Russia. Russian socialists' major contribution to the "sex question" during the nineteenth century was Nikolai Chernyshevsky's canonical novel of ideas, *What Is to Be Done?* (1863).[3] The book depicted "new people" who subordinated their heterosexual but otherwise unconventional love lives to their social activism. Chernyshevsky laid out a scheme that celebrated women's self-determination through "fictional marriage," a device that liberated women of the intelligentsia from patriarchal oppression. His scheme emphasized the virtue of passionate, romantic love, wherever it might be found, over sterile devotion to matrimonial duty; it also trumpeted a radical rejection of jealousy as bourgeois and proprietorial. Yet this code of "free love" was itself subject to the needs of a movement for social and political change. These values were an integral feature of the radical socialist movement's critiques of middle-class morality by the time revolution came in 1917. An end to the strictures of patriarchy in marriage, a "free love" liberated from the exactions of property and religion, and a demanding cult of sublimation for the good of the coming revolution were commonplaces of socialist discourse, which sought a new morality based on collective principles.[4]

Of course, the ethos of deferral, sublimation, and subordination of personal pleasures and satisfactions to a "higher" purpose was neither originally nor exclusively socialist in Russian culture. Conventional

morality, as propounded for centuries by the Orthodox Church, fostered deferral and sublimation as religious virtues. Taking this ethos to its extreme in 1891 in his controversial short story "The Kreutzer Sonata," Leo Tolstoy offered a bleak vision of sexual life as a destructive force, and proposed abstinence as the only moral alternative.[5] Silver Age novelists and poets celebrated Eros not as a goal in itself but as a path to transcendence, to the construction of a new spiritual life as secular utopia.[6] After the lifting of censorship in 1905, socialists were not alone in finding the commercialization of sex repugnant in boulevard literature and the new medium of the cinema.[7] By the last years of the imperial regime, Russians of many political and intellectual currents viewed sex as a powerful natural force that required channeling for civic and spiritual well-being. Sex, in the words of Eric Naiman, had become an "ideological signpost" for the Russian intelligentsia, a field of such significance that every thinker felt the need to incorporate judgments about it into his or her worldview.[8] Intellectuals and social critics deplored an apparent flourishing of "petty-bourgeois philistinism," in Russian summed up in the evocative word *meshchanstvo*. Where sex was under the microscope, the pathologies of *meshchanstvo* included patriarchy, jealousy, property ties, religious prejudice, ignorance about disease, prudery about the body, pleasure for its own sake, sexual "perversion" (an elastic classification), and the double standard. The flaying of the petty bourgeoisie for its sexual sins, real and ascribed, continued well into the Soviet period.[9]

Russian Marxists unhesitatingly accepted the authority of medical science to serve as a guide to the healthy direction of sexuality under socialism. Medicine offered a language of modernity that, for Bolsheviks, drained this controversial issue of "bourgeois philistinism" and satisfied the predominantly ascetic instincts of the party. For all their protests that they were not prudes, Soviet leaders of Lenin's generation—and many younger Bolshevik enthusiasts—were displeased by the lively popular interest in sexual matters. The prevailing view in the party, inherited from Russian radicalism, was that individual fulfillment must wait until the revolution was secured and socialism developed. Reasoning nevertheless that nature was not to be denied, the leadership adopted radical socialist laws on women and the family to channel sex rationally into what were presumed to be its natural grooves, undistorted by bourgeois values, including the distortions imposed by property considerations and religious morality.[10] Medicine would be the key to most sex questions previously framed as moral problems in tsarist regulation.

Medicalization crucially transferred authority in sexual matters from the individual to the collective, to be presided over by its agents, physicians, and specialists in sexology. This "rationalizing" current in

treatment regimes of the 1950s–1960s, demonstrating a "modern" grasp of the question. Simultaneously, in their aspirations and operations Soviet doctors treating hermaphrodites showed how a precociously modern medical approach could reinscribe and stabilize a two-sex gender order.[26]

By institutionalizing "objective" scientific investigation in the Soviet judicial and administrative system, Bolsheviks hoped for a modern, secular, and rational approach to sexual disorder. Medicine would replace religion and bourgeois moralizing as a source of authority in these often intractable cases. In this aspiration, Bolsheviks and their legal and medical fellow travelers adhered to a common trend toward secularization and medicalization in the treatment of sexual questions in the industrialized world. During and after the upheavals of the First World War, medical and public health officials bid for and acquired custody of many sex-related issues; in the nonsocialist world sex reformers and sexologists presented themselves as enemies of secretive Victorian morality, a prudish silence to be overcome by propounding sexual hygiene and enlightenment. Many sex reformers pointed to Soviet lawmaking and practices as leading the way to sexual modernity. The Bolsheviks did not single-handedly invent "sexual revolution" but tried to harness it to the requirements of a peculiarly Soviet, socialist regime.[27]

Looking for a Sexual Revolution

The history of the Soviet sexual revolution has primarily been told by examining what people said rather than what people did. Problems of access to sources tended to determine this focus; however taboos on the topic of sexuality during the Cold War also inhibited scholarship on both sides of the iron curtain. Historians of Russia's women began in the 1970s to explore the record of the sexual revolution through the lenses of women's liberation and later gender studies. The emphasis was on a recuperative project that revealed for a generation increasingly skeptical about the totalizing claims of Marxism what traditional historical and political narratives of revolution had trivialized: the role of women's emancipation and the women's question in Soviet history.[28] Scholars more recently have turned to literary and journalistic sources to illuminate, often with great insight, what commissars and critics thought and said about the Soviet sexual revolution. The NEP era was one of a plurality of views, demonstrable from debates in party journals, from literary works produced by non-Communist and Communist authors, from the sex-themed pamphlets and bestsellers of medical experts and the advice columns of newspapers. Censorship was inefficient, and commercial considerations in the market economy conditions encouraged editors to publish sex-themed material.[29] For the Communist regime it was, as Eric

Naiman and Frances Bernstein have argued, a time of considerable anxiety about authority that found vivid expression in literary, journalistic, and medical debates about sexuality.[30]

This book looks at the record of practices to enrich the narrative of sexual revolution. As Bernstein has shown in her study of early Bolshevik sex advice for the masses, the practices of medicine as documented in newly accessible archival holdings are a crucial source for understanding how the sexual revolution was to be disciplined immediately from its inception.[31] Her work shows how Soviet doctors imagined and encouraged positive sexual behavior that conformed to the dominant, essentializing understanding of revolutionary sexual values; in contrast, this book examines medical practices when negative behavior or conditions were encountered. Doctors' actions as forensic experts and clinicians were part of a wider range of "the repeated, routine practices of everyday life" that determined how the socialist revolution was made a reality.[32] The medical practices of these doctors were knitted into a larger, international process of transformation. Soviet sexology in its methodologies and outlook was part of a European-wide trend toward the state exploitation of "biopower," and in imposing an apparently "revolutionary," "rational" framework on human sexuality the Soviet version of sexual science did much to constrain it in new channels.[33] Work on homosexuality, prostitution, and infanticide has demonstrated the degree to which the Bolshevik interpretations of socialist sexual traditions, and policies of "sexual revolution," were formed by the science and medicine of the 1920s.[34] Few works have yet made a systematic examination of the grassroots practices of Bolshevik justice and administration when handling sexual disorder.[35]

In English-language scholarship, critical histories of forensic medicine are surprisingly underdeveloped (especially if one considers the current boom in forensics-based television dramas, which ought to stimulate a new wave of interest inside and beyond the academy).[36] Recent works explore the origins and struggle for professional recognition of forensic expertise in the United Kingdom and the United States. Much of this work focuses on the gradually increasing role of the psychiatrist in crime investigation and prosecution. Other significant areas of interest include the cultural transmission of modern forensic science from its origins in revolutionary France of the late eighteenth century to other nations, and the evolution of forensic laboratories and services to police investigators in various jurisdictions.[37] Continental historiography similarly focuses on issues of professionalization and institution-building.[38] Russian-language histories of tsarist and Soviet forensic medicine provide a fund of knowledge for the critical scholar but do not depart from an uncritical narrative of progress.[39] In English, imperial Russian forensic medicine has

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Dan Healey provides the first historical ex i-
cine's entanglement in matters concerni d
dentity, exposes a range of scientific pract y,
and offers glimpses of ordinary people who had to navigate the medico-legal terrain being mapped out by the experts."
—Bulletin of the History of Medicine

With its clear arguments and excellent sections providing context for both Russia and the world, this book is a must for those interested in the revolutionary era and would be a superb addition to undergraduate and graduate courses in gender, medical, and Soviet history."
—Slavic Review

Healey's book contains much that will be of interest to historians of sexuality."
—American Historical Review

Bolshevik Sexual Forensics explores the institutional history of Russian and Soviet forensic medicine and examines the effects of its authority when confronting sexual disorder. Dan Healey compares sex crime investigations from Petrograd and Sverdlovsk in the 1920s to the numerous publications by forensic doctors and psychiatrists of the prerevolutionary and early Soviet periods to illustrate the role that these specialists played. In addition, Healey presents a fascinating look at how doctors diagnosed and treated hermaphroditism, showing how Soviet physicians revolutionized the standard scientific view in these cases by taking into account individual desire. *Bolshevik Sexual Forensics* sheds light on unexplored radical and reactionary forces that shaped the Bolshevik "sexual revolution" as lawmakers defined new ways of seeing sexual crime and disorder. This groundbreaking book will appeal to Soviet historians and those interested in gender studies, sexuality, medicine, and forensics.

DAN HEALEY is a Professor of Modern Russian History at St Antony's College, University of Oxford. He is the author of *Russian Homophobia from Stalin to Sochi.*

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Neuroplasticity and Rehabilitation



edited by

SARAH A. RASKIN

Neuroplasticity and Rehabilitation

edited by **Sarah A. Raskin**

Brain plasticity is the focus of a growing bo of research with significant implications f neurorehabilitation. This state-of-the-art vo ume explores how knowledge about plasticity being put to use toward developing and evalua ing innovative intervention approaches. Leade in the field describe ways in which brain-injur individuals may be helped not only to compe sate for their loss of cognitive abilities, but al possibly to restore those abilities.

The book examines the extent to which dar aged cortical regions can actually recover ar resume previous functions, as well as how inta regions are recruited to take on tasks once mec ated by the damaged region. Evidence-bas rehabilitation approaches are presented for range of impairments and clinical populatior including both children and adults. Contrib tors also address how they are using neuroima ing techniques to measure brain changes relat to treatment. Chapters review the nuts and bo of treating patients with:

- Motor impairments.
- Dyslexia.
- Attention deficits.
- Aphasia.
- Executive function deficits.
- Learning and memory problems.

The concluding chapter discusses pharmac logical therapies that are used in cognitive r habilitation, emphasizing that much remains be learned about the impact of medications neuroplasticity. Throughout the volume, futu directions for improving clinical research ar patient care are highlighted.

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Neuroplasticity and Rehabilitation

"Raskin has assembled an impressive team of experts who combine critical knowledge of brain reorganization and repair with applied rehabilitation science, providing a long-overdue conceptual integration. This thought-provoking book is a 'must read' for students; for scholars, who may have expertise in the basic science but not in clinical application; and for clinicians. The book is truly translational in nature."—John DeLuca, PhD, ABPP, Professor of Physical Medicine and Rehabilitation, UMDNJ–New Jersey Medical School; Vice President for Research, Kessler Foundation, West Orange, New Jersey

"Raskin and her colleagues point out that the same mechanisms underlying experience-based neural plasticity also contribute to recovery of function after brain damage, and demonstrate how this informs rehabilitation research and practice. I can envision this book serving as a standard text for the next generation of researchers and clinicians hoping to address the restoration of cerebral function." —Keith D. Cicerone, PhD, ABPP-CN, Director of Neuropsychology, JFK Johnson Rehabilitation Institute, Edison, New Jersey

"The take-home message of *Neuroplasticity and Rehabilitation* is not only that our brains are more plastic than we ever expected, but also that we are learning how to influence this plasticity through informed treatment to achieve truly meaningful rehabilitation outcomes. From animal studies to sophisticated human trials, the book gathers a wealth of information from eminent experts. This book is absolutely essential reading for basic neuroscientists as well as for rehabilitation professionals at all levels of care. Understanding how and why the study of neuroplasticity can inform treatment choices will allow clinicians to make cutting-edge clinical decisions....The future of rehabilitation has now been linked inextricably to the growing field of neuroplasticity." —Jill Winegardner, PhD, lead psychologist, Oliver Zangwill Centre for Neuropsychological Rehabilitation, United Kingdom

"This is a comprehensive book incorporating the latest research on neuroplasticity in diverse areas of cognitive function. It serves as an authoritative guide to how clinical practice can effectively incorporate significant new findings about brain plasticity and learning. It will be equally appropriate and useful for graduate students, experienced clinicians, and researchers in the field." —Wayne A. Gordon, PhD, Jack Nash Professor, Department of Rehabilitation Medicine, Mount Sinai School of Medicine, New York

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